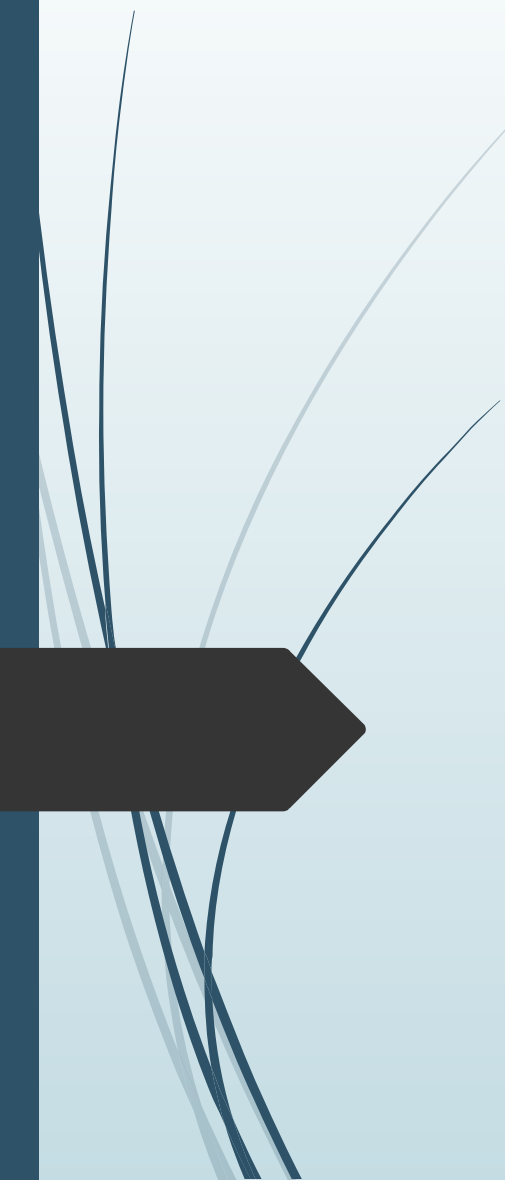


Introduction to Artificial Intelligence

Week 11

Traveling Salesman Problem (TSP)

- "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?"

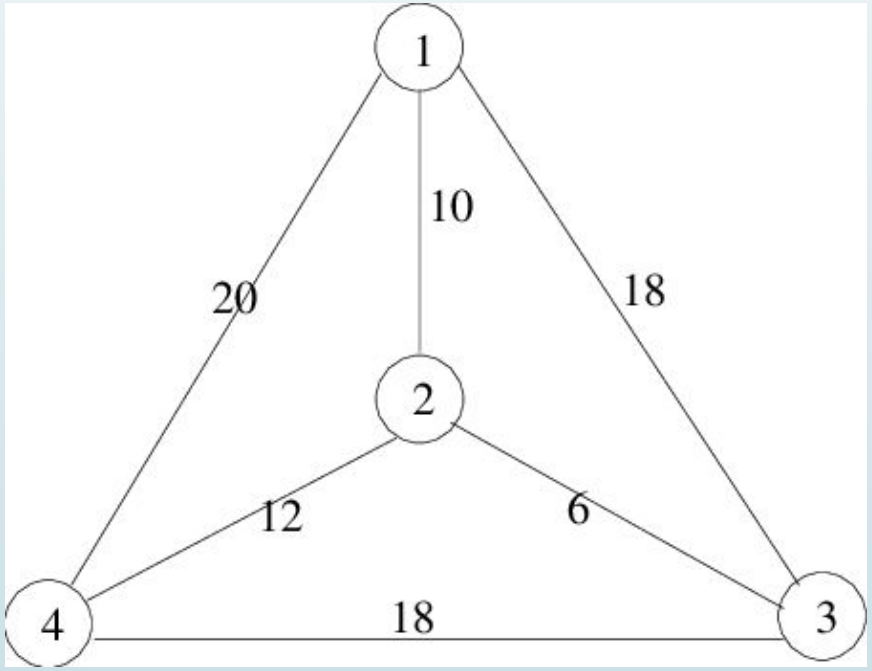


Traveling Salesman Problem (2)

	C1	C2	C3	C4	C5
C1	0	132	217	164	158
C2	132	0	290	201	79
C3	217	290	0	113	303
C4	164	201	113	0	196
C5	158	79	303	196	0

The solution can be written as C2->C5->C1->C4->C3->C2. An itinerary that begins and ends at the same city and visits each city once is called a tour.

TSP – Graph Representation



Ant Colony Optimization (ACO)

- The study of artificial systems modeled after the behavior of real ant colonies and are useful in solving discrete optimization problems
- Introduced in 1992 by Marco Dorigo
- It is a population-based metaheuristic used to find approximate solutions to difficult optimization problems
- https://www.youtube.com/watch?v=qfeymoF8pb4

Artificial Ants

- A set of software agents
- Stochastic
- Based on the pheromone model
  - Pheromones are used by real ants to mark paths. Ants follow these paths (i.e., trail-following behaviors)
- Incrementally build solutions by moving on a graph
- Constraints of the problem are built into the heuristics of the ants

A key concept: Stigmergy

Stigmergy is:

indirect communication via interaction with the environment.

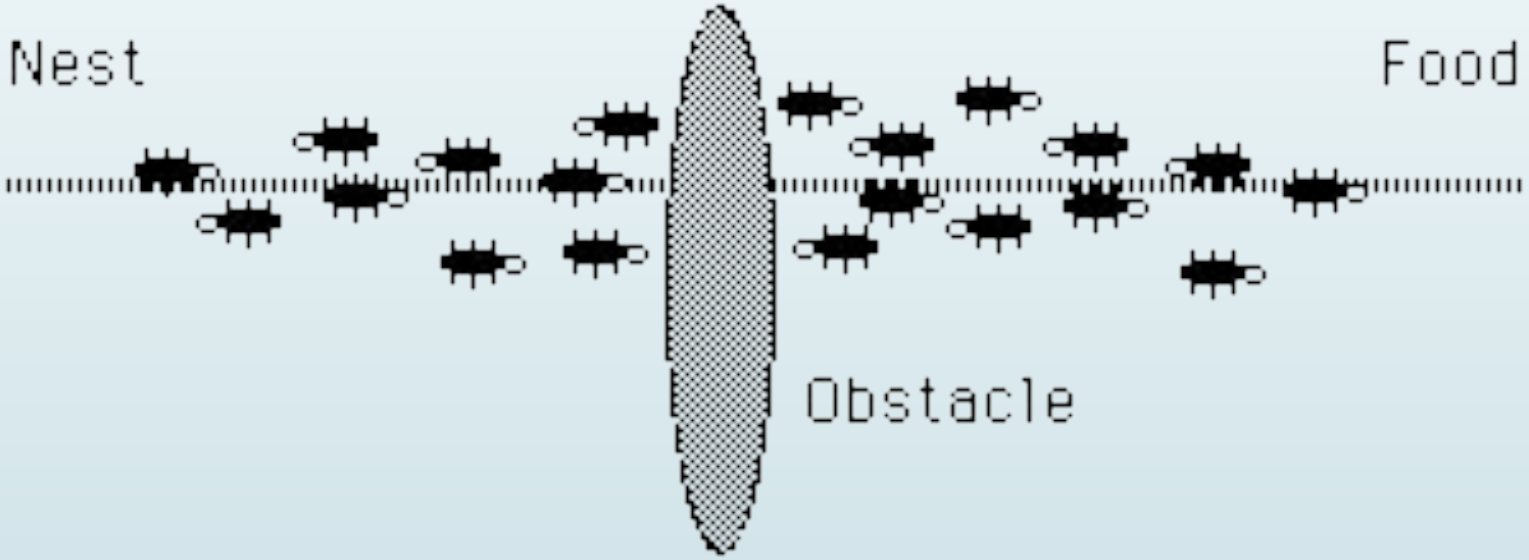
- A problem gets solved bit by bit ..
- Individuals communicate with each other in the above way, affecting what each other does on the task.
- Individuals leave *markers* or *messages* – these don't solve the problem in themselves, but they affect other individuals in a way that helps them solve the problem …

Naturally Observed Ant Behavior



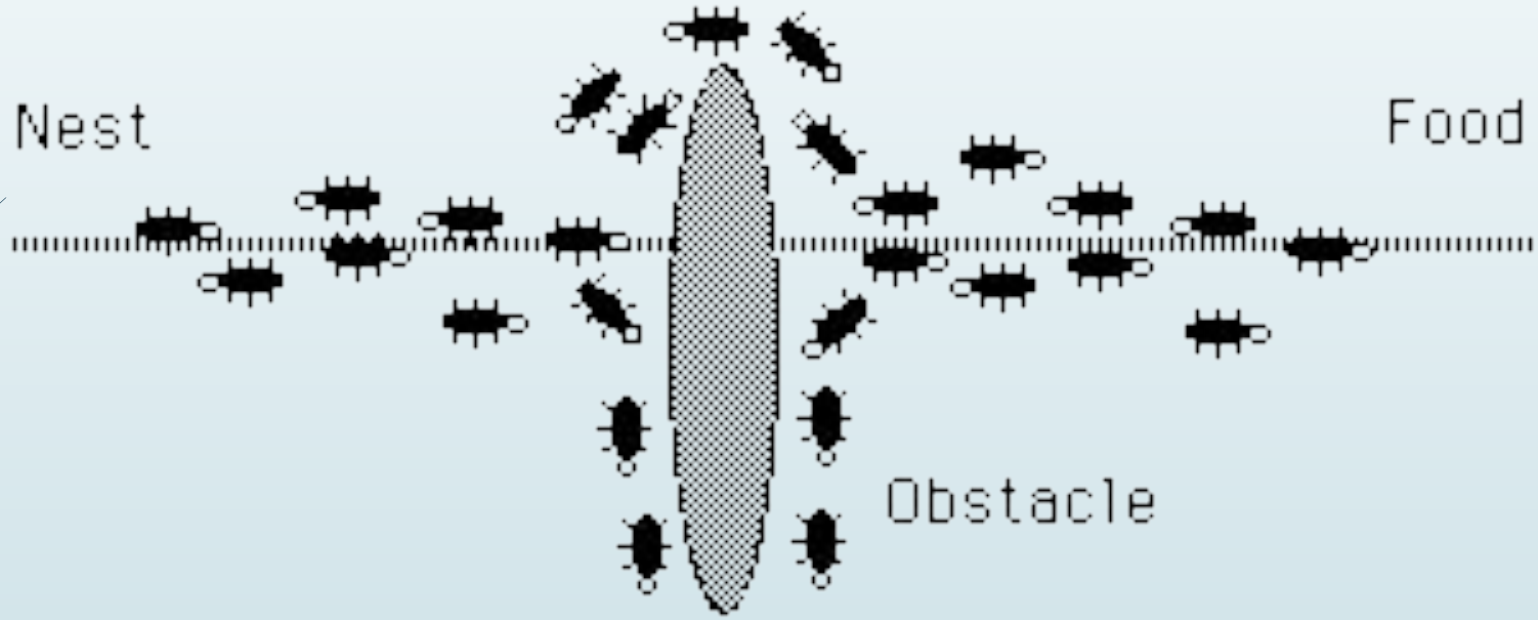
All is well in the world of the ant.

Naturally Observed Ant Behavior



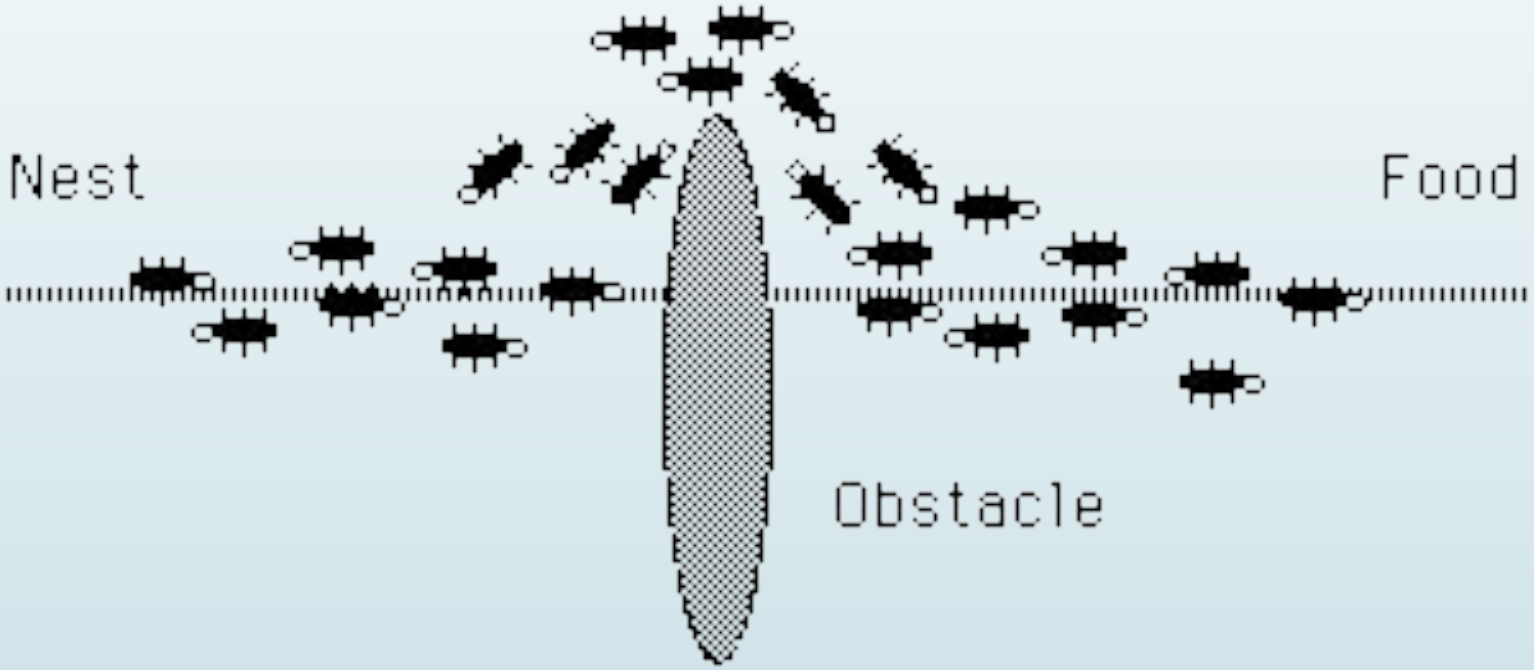
Oh no! An obstacle has blocked our path!

Naturally Observed Ant Behavior



Where do we go? Everybody, flip a coin.

Naturally Observed Ant Behavior



Shorter path reinforced.



Stigmergy in Ants

Ants are behaviorally unsophisticated, but collectively they can perform complex tasks.

Ants have *highly developed sophisticated sign-based stigmergy*

- They communicate using pheromones;
- They **lay trails of pheromone** that can be followed by other ants.

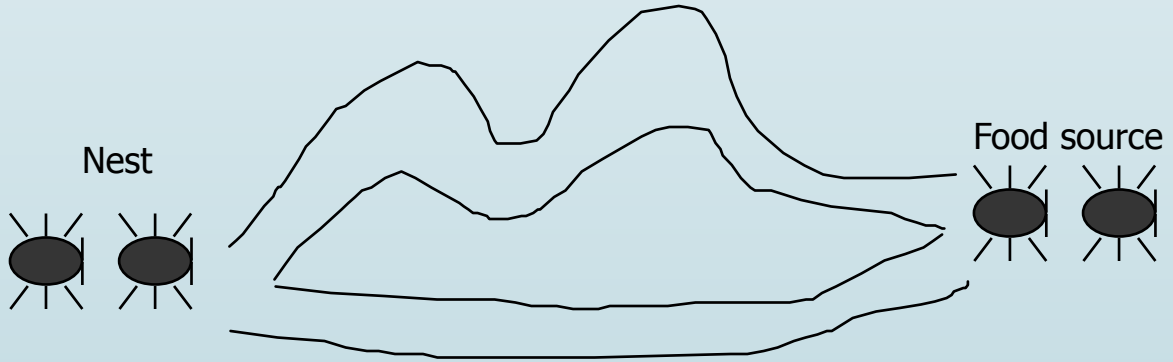
- If an ant has a **choice of two pheromone trails** to follow, one to the NW, one to the NE, but the NW one is *stronger* – which one will it follow?

Pheromone Trails

Individual ants lay pheromone trails while travelling from the nest, to the nest or possibly in both directions.

The pheromone trail gradually evaporates over time.

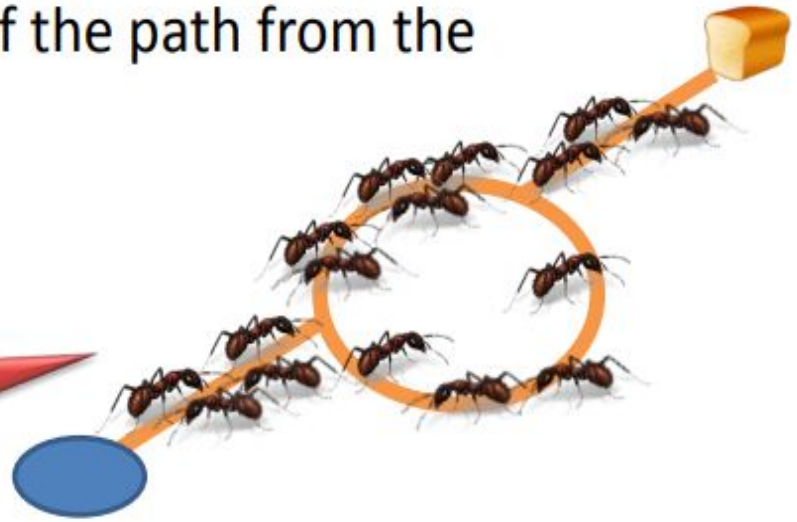
But pheromone trail strength accumulate with multiple ants using path.



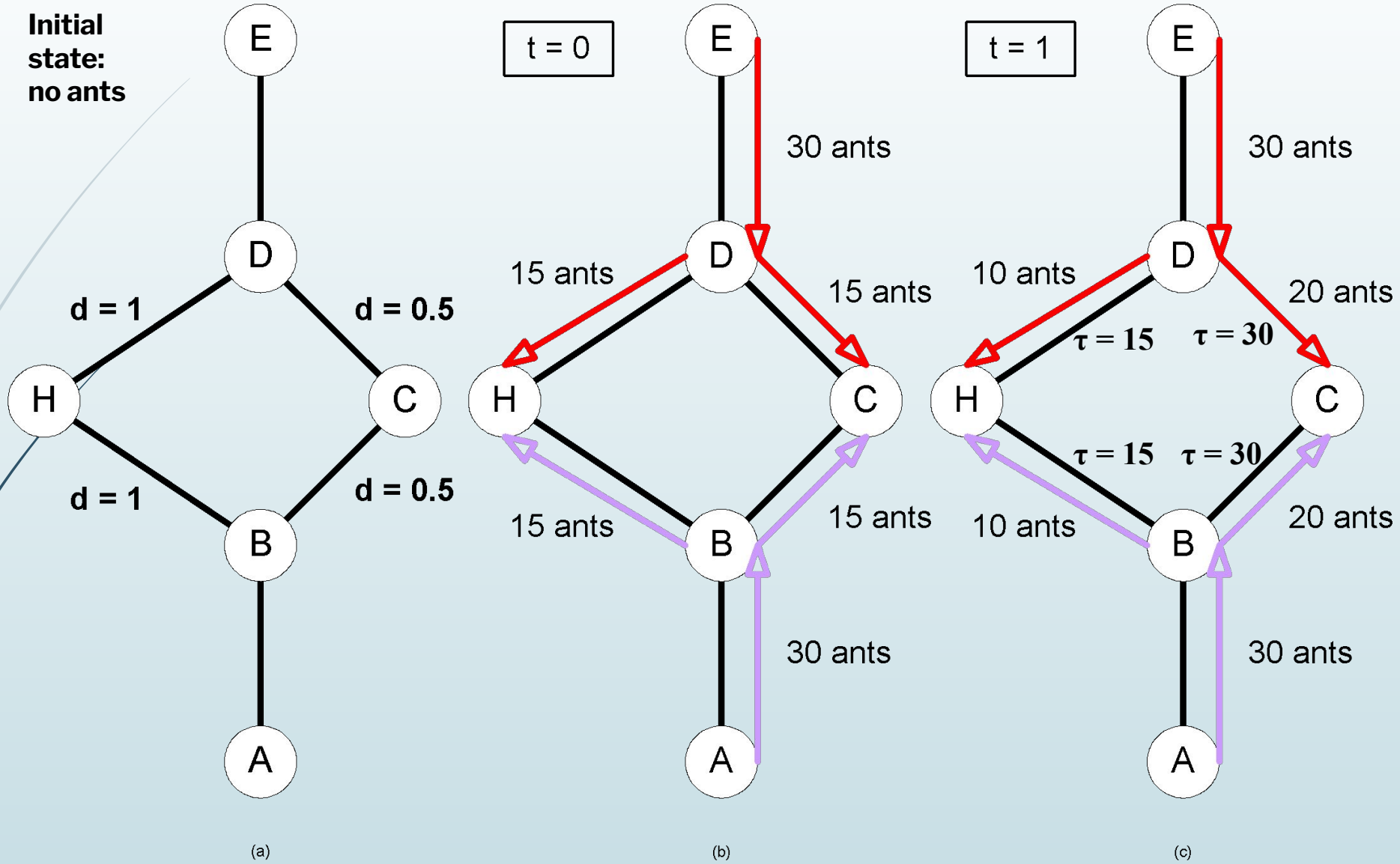
- Properties of the pheromone:
  - The pheromone is olfactive and volatile.
  - The pheromone is stronger if more ants go along the same path (reinforced by number).
  - The pheromone is stronger if the path from the nest to the food is shorter

    (less evaporation).

After a while, the shortest path will be found automatically!



Pheromone Trails continued



ACO Algorithm

Set all parameters and initialize the pheromone trails

Loop

Sub-Loop

Construct solutions based on the state transition rule

Apply the online pheromone update rule

Continue until all ants have been generated

Apply Local Search

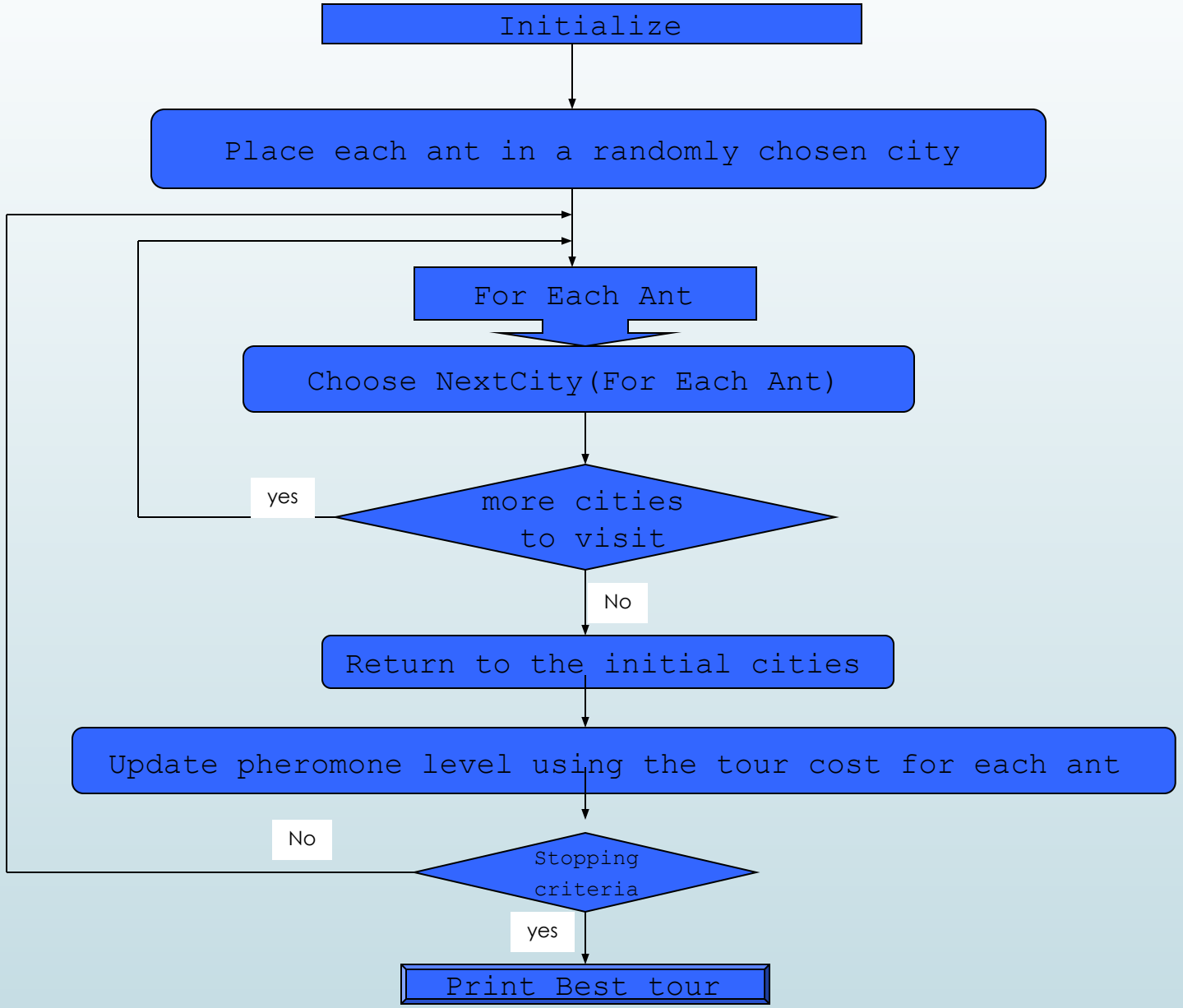
Evaluate all solutions and record the best solution so far

Apply the offline pheromone update rule

Continue until the stopping criterion is reached

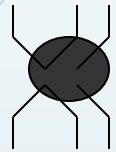


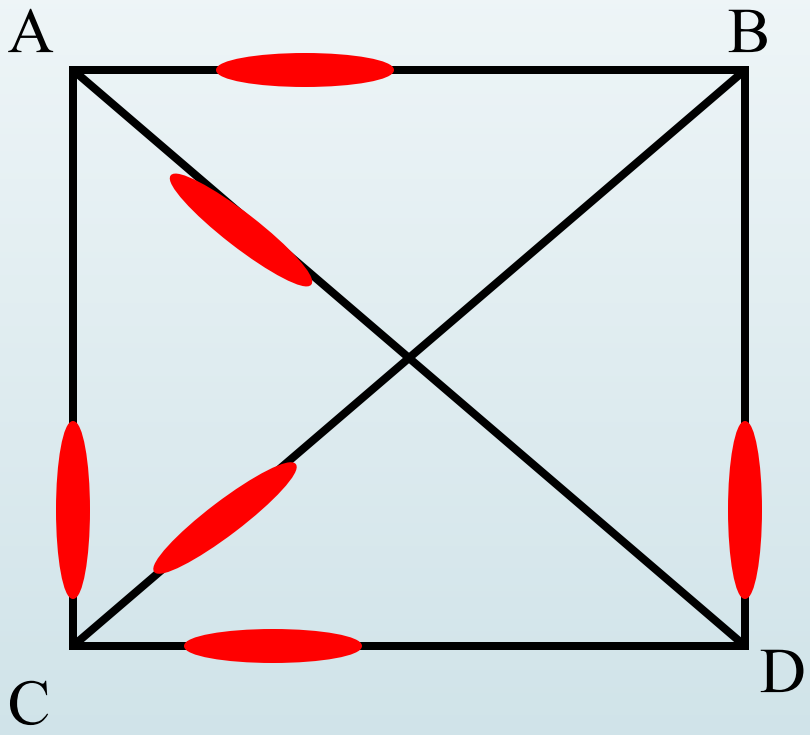
Ant Systems Algorithm for TSP



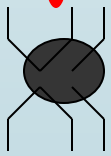
E.g. A 4-city TSP

Initially, random levels of pheromone are scattered on the edges





Pheromone

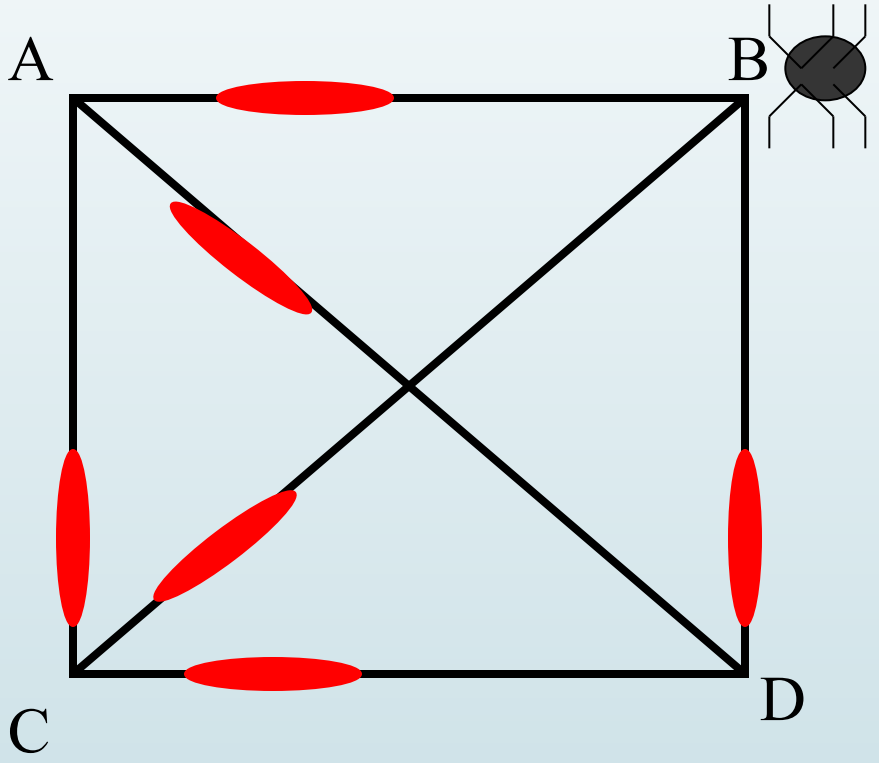


Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

An ant is placed at a random node



Pheromone

Ant

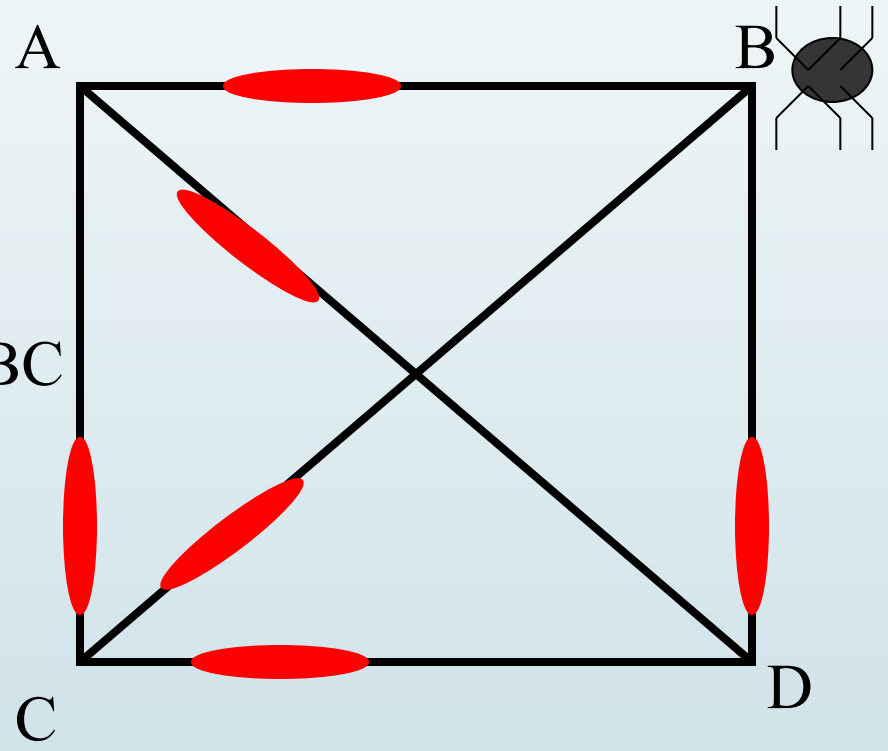
AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

The ant decides where to go from that node, based on probabilities calculated from:

- pheromone strengths,
- next-hop distances.

Suppose this one chooses BC



Pheromone

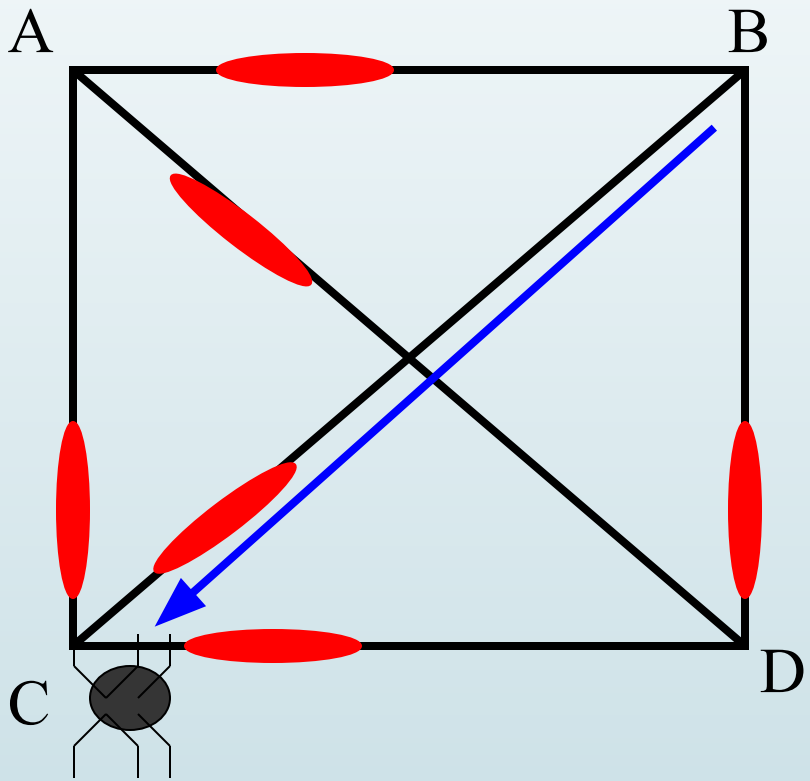
Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

The ant is now at C, and has a `tour memory' = {B, C} – so he cannot visit B or C again.

Again, he decides next hop (from those allowed) based on pheromone strength and distance;
suppose he chooses CD



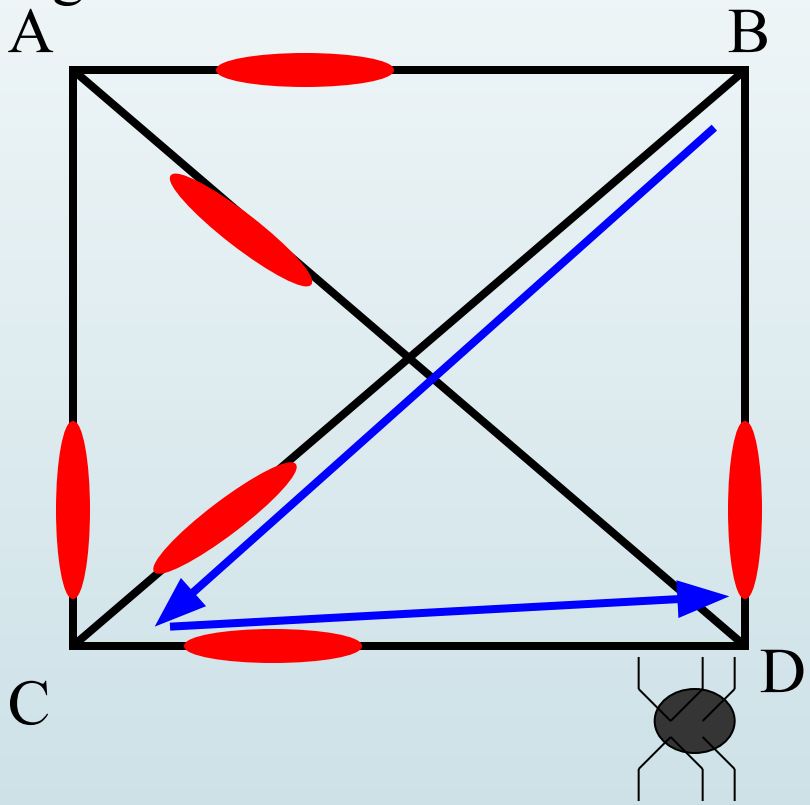
Pheromone

Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

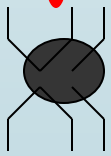
E.g. A 4-city TSP

The ant is now at D, and has a `tour memory' = {B, C, D}
There is only one place he can go now:





Pheromone

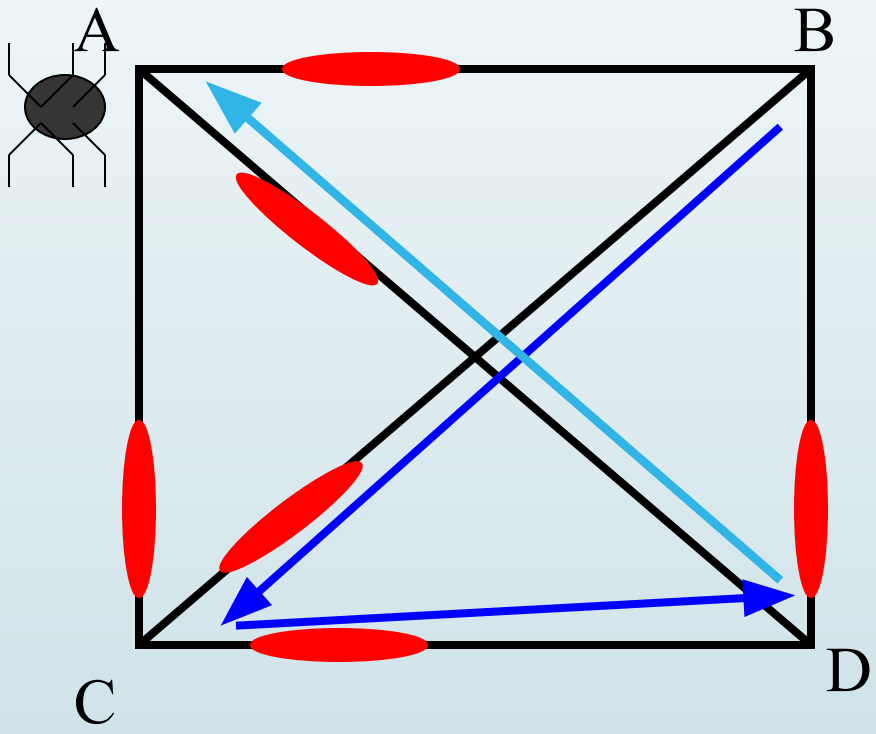


Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

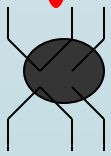
E.g. A 4-city TSP

So, he has nearly finished his tour, having gone over the links: BC, CD, and DA.





Pheromone



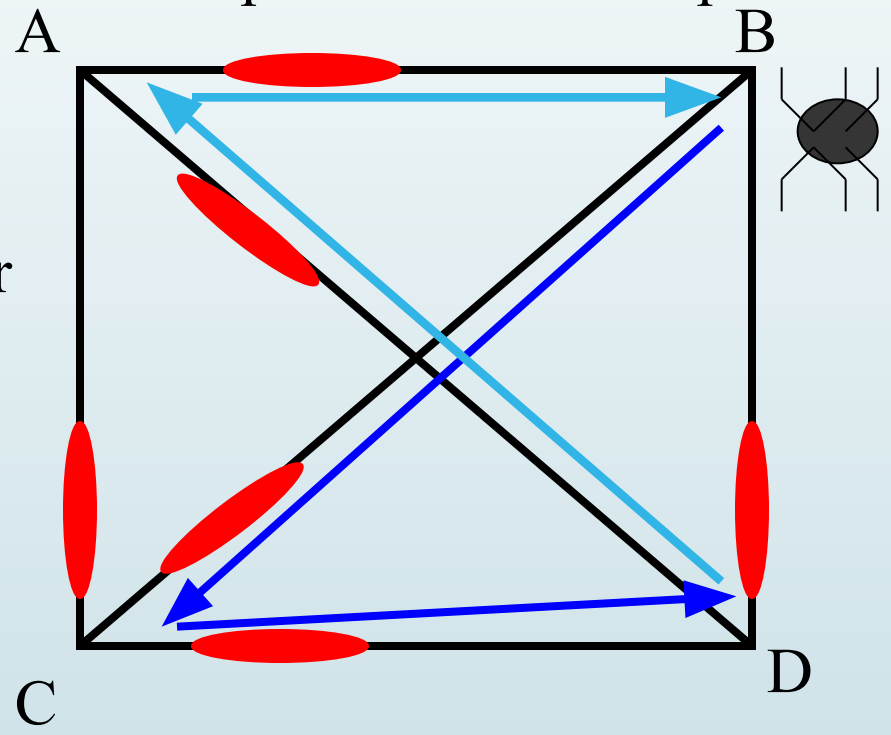
Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

So, he has nearly finished his tour, having gone over the links: BC, CD, and DA. AB is added to complete the round trip.

Now, pheromone on the tour is increased, in line with the fitness of that tour.



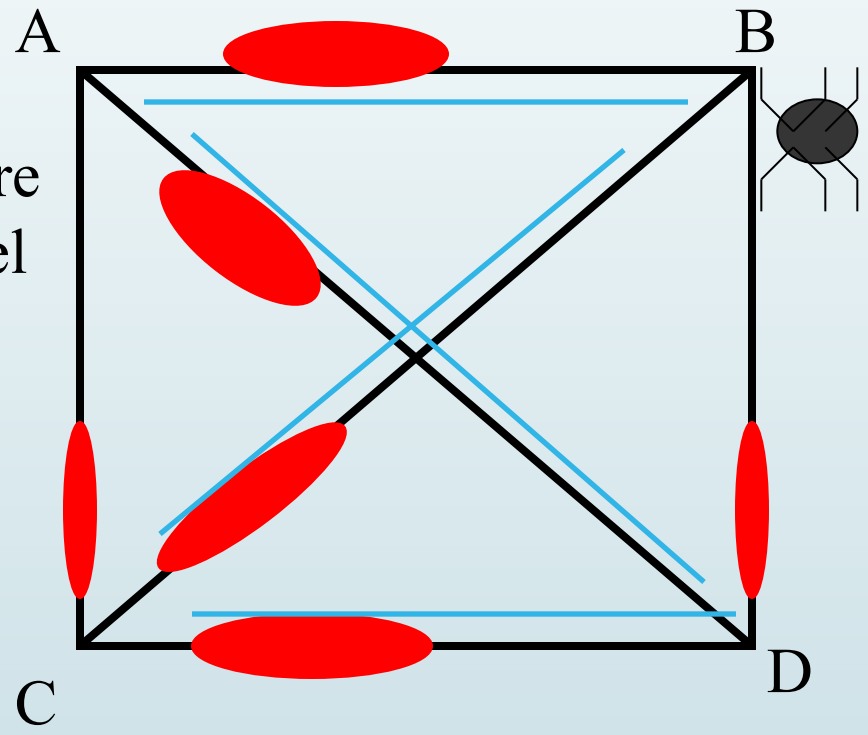
Pheromone

Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

Next, pheromone everywhere is decreased a little, to model decay of trail strength over time



Pheromone

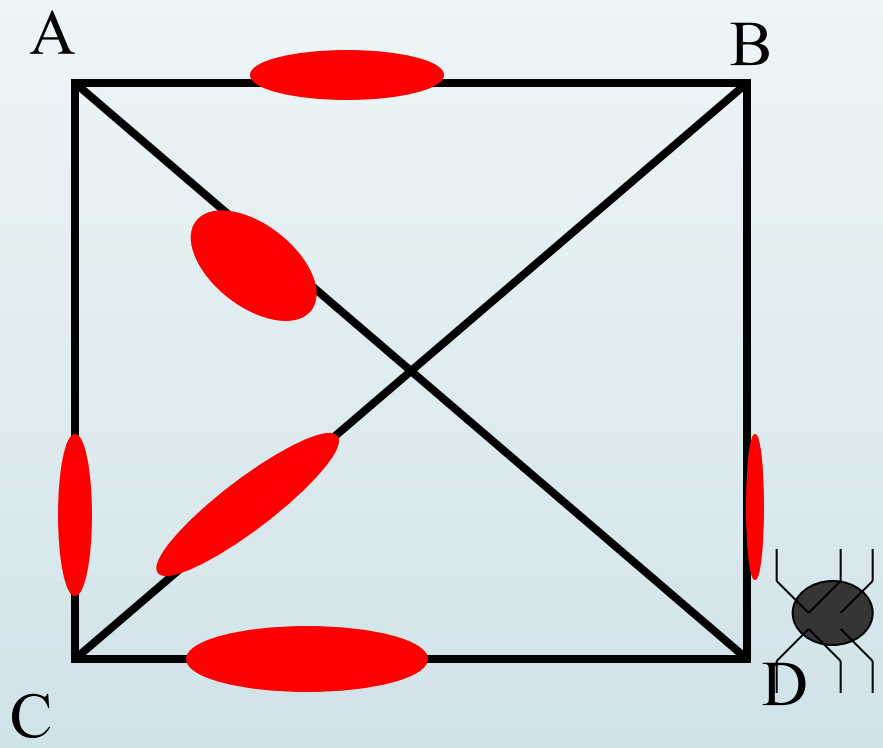
Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

We start again, with another ant in a random position.

Where will he go?



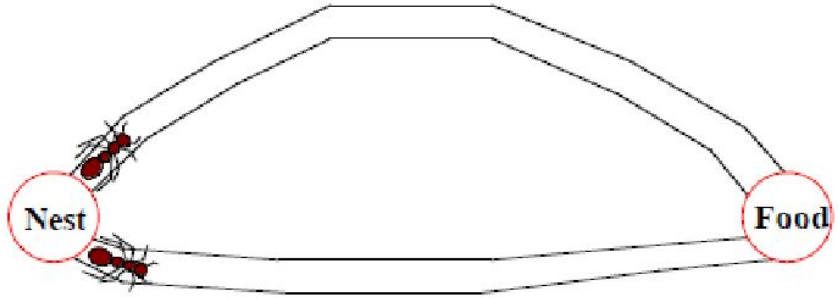
Pheromone

Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

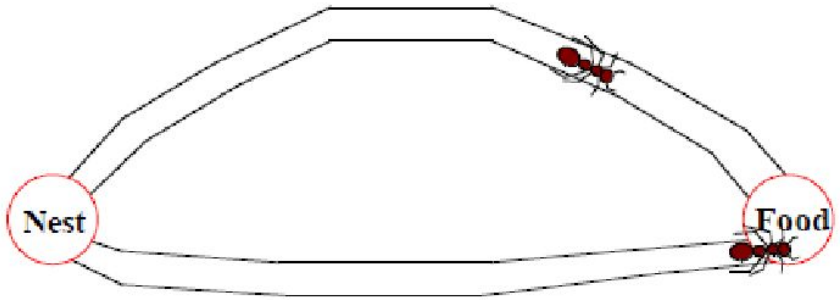
Ant foraging

Cooperative search by pheromone trails



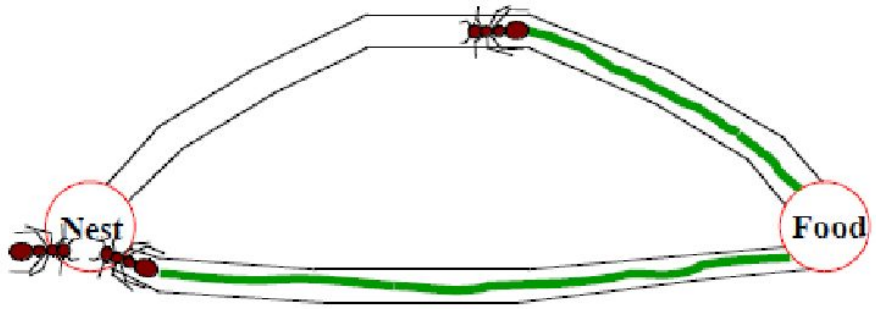
Ant foraging

Cooperative search by pheromone trails



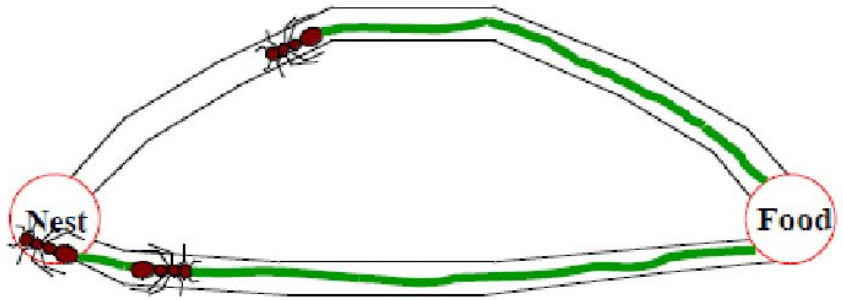
Ant foraging

Cooperative search by pheromone trails



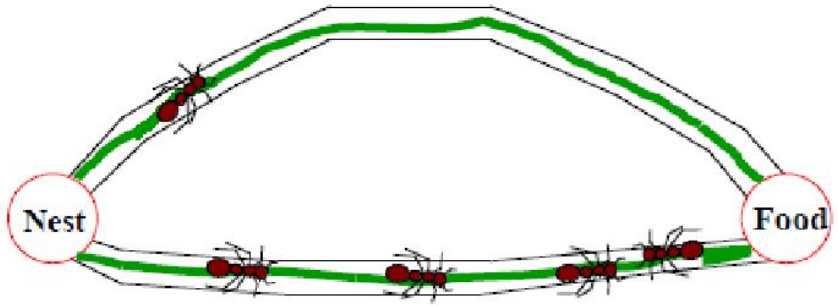
Ant foraging

Cooperative search by pheromone trails



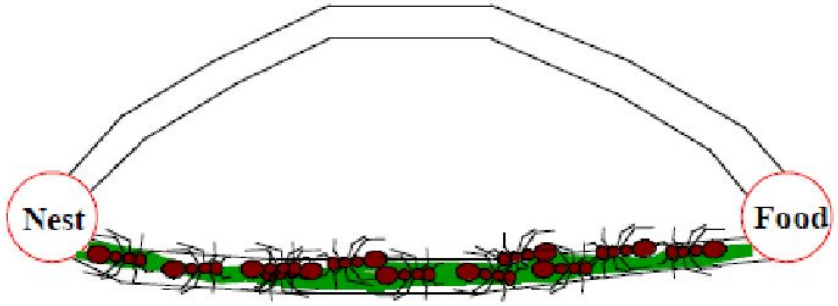
Ant foraging

Cooperative search by pheromone trails



Ant foraging

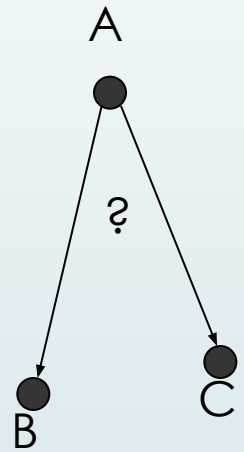
Cooperative search by pheromone trails



Ant Colony Optimization

Characteristics of the algorithm

The probability of moving from Point A to Point B depends on the attractiveness of the move and the trail level.



Attractiveness - an *a priori* desirability (typically based on the inverse distance) of the move.

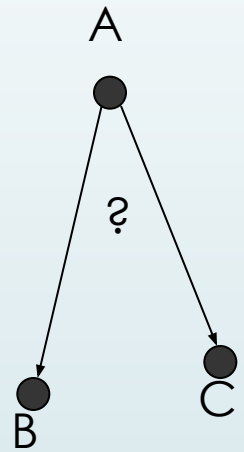
Trail level - an *a posteriori*, computed attractiveness. This is computed after an ant completes a solution, increasing or decreasing the desirabilities of components of a solution.

$$\text{Prob}_{A\to B} = \left(\frac{(\text{Amount of pheromone}_{A\to B})(\text{Desirability of state transition}_{A\to B})}{\sum(\text{Amounts of pheromones})(\text{Desirabilities})} \right)$$

Ant Colony Optimization

Characteristics of the algorithm

Amounts of pheromones are updated after a solution has been generated.



(1-*p*) prior pheromone + Δpheromone

Where:
***p* = a coefficient related to the evaporation rate.**

if A→B path is used, pheromone levels increase

$$\Delta\text{pheromone} = Q / L_k$$

Where:
Q = a constant
L_k = solution length

Else pheromone levels decrease, since in this case

Δpheromone = 0

Ant Colony Optimization

Necessary parameters

1) The initial population of ants (an ant cycle).

2) The total number of tours.

3) The evaporation rate.

4) The amount of pheromone applied to a section of a cycle or solution.

5) The constant Q.

The transition rule

τ_{ij} is the amount of pheromone currently on the path that goes directly from city *i* to city *j*.

η_{ij} is the heuristic value of this link – in the classic TSP application, this is chosen to be 1/distance(*i*,*j*) -- i.e. the shorter the distance, the higher the heuristic value.

$p_k(i,j)$ is the probability that ant *k* will choose the link that goes from *i* to *j*

State Transition Probability:

$$P^k{}_{ij} = \frac{(\tau_{ij})^\alpha (\eta_{ij})^\beta}{\sum_{l \in U} (\tau_{il})^\alpha (\eta_{il})^\beta}$$

unvisited cities *SetU*

Where our ant is at city *i*, and *j* is a city as yet unvisited on its tour, and the summation is over all of *k*'s unvisited cities

Global pheromone update

$\Delta\tau_{ij}^{k}$ is amount of pheromone added to the (i, j) link by ant k.

NA is the number of ants

ρ is a parameter called the pheromone decay rate.

L_k is the length of the tour completed by ant k

τ_{ij}^{k} at the next iteration becomes:

Pheromone evaporation

$$\tau_{ij} \leftarrow (1-\rho)\tau_{ij}, \quad \forall (i,j) \in L$$

Where

$$\tau_{ij}^{new} = \rho\tau_{ij}^{old} + \sum_{k=1}^{NA} \Delta\tau_{ij}^{k}$$

$$\Delta\tau_{ij}^{k} = Q, \quad \Delta\tau_{ij}^{k} = \frac{Q}{d_{ij}} \quad or \quad \Delta\tau_{ij}^{k} = \frac{Q}{L^{k}}$$

DEMO (PYTHON)

Questions?